AWS VPC NETWORKING PROJECT REPORT

Virtual Private Cloud, Subnets, Route Tables, EC2 Instances and Private Network Connectivity Testing

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Prepared from the AWS console screenshots and connectivity-test output supplied for this project.

1. Project Overview

The aim of this project is to explore AWS virtual networking and verify communication between EC2 instances using private IPv4 addresses. The work shown in the supplied console screenshots includes VPCs, subnets, route tables, three running EC2 instances, security-group inbound rules, and ping-based connectivity checks.

2. Objectives

• Review VPCs and subnet resources in the AWS VPC console.
• Inspect route tables and their subnet associations.
• Launch and verify EC2 instances used for the networking exercise.
• Configure inbound security-group rules for SSH and ICMP testing.
• Test reachability between the two application instances using private IP addresses.

3. Resources Observed

Resource	Details visible in screenshots
VPCs	ewb-vpc-a, ewb-vpc-b, and one unnamed VPC; all displayed as Available.
Subnets	Six subnets are indicated in the console. Visible names include ewb-subnet-a-public and ewb-subnet-b-private.
Route tables	Seven route tables are indicated. Visible names include ewb-rt-a-public, ewb-rt-a-private, and ewb-rt-b-private.
EC2 instances	ewb-bastion, ewb-app-a, and ewb-ap-b; all shown as Running with 3/3 status checks passed.
Application private IPs	ewb-ap-b: 172.31.45.121; ewb-app-a: 172.31.32.173.

Note: The screenshots do not show every CIDR block, route destination, or VPC-peering status. This report therefore describes only details visible in the supplied evidence.

4. AWS Console Evidence

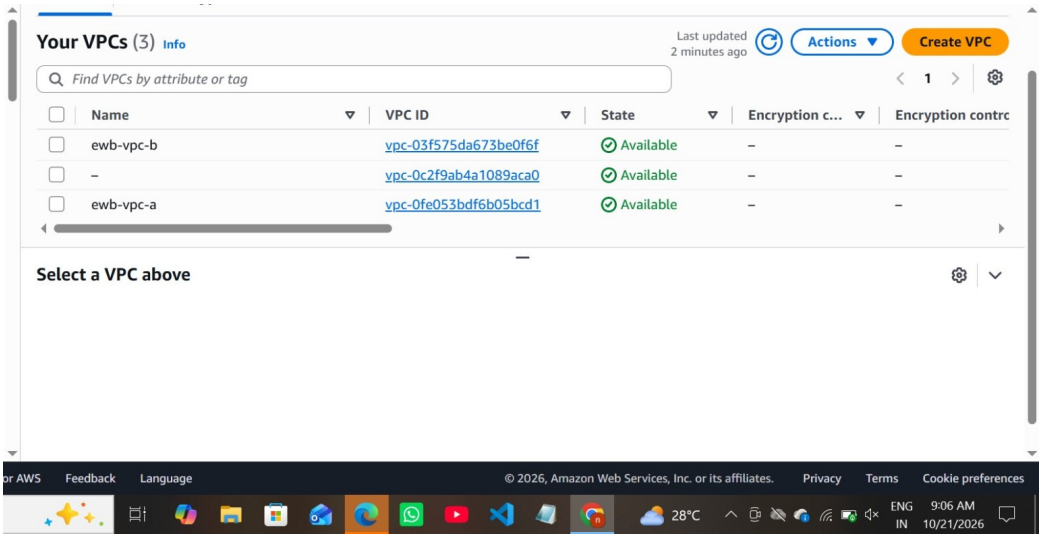


Figure 1. VPC inventory in the AWS VPC console. The account shows ewb-vpc-a, ewb-vpc-b, and an unnamed VPC; all are shown as Available.

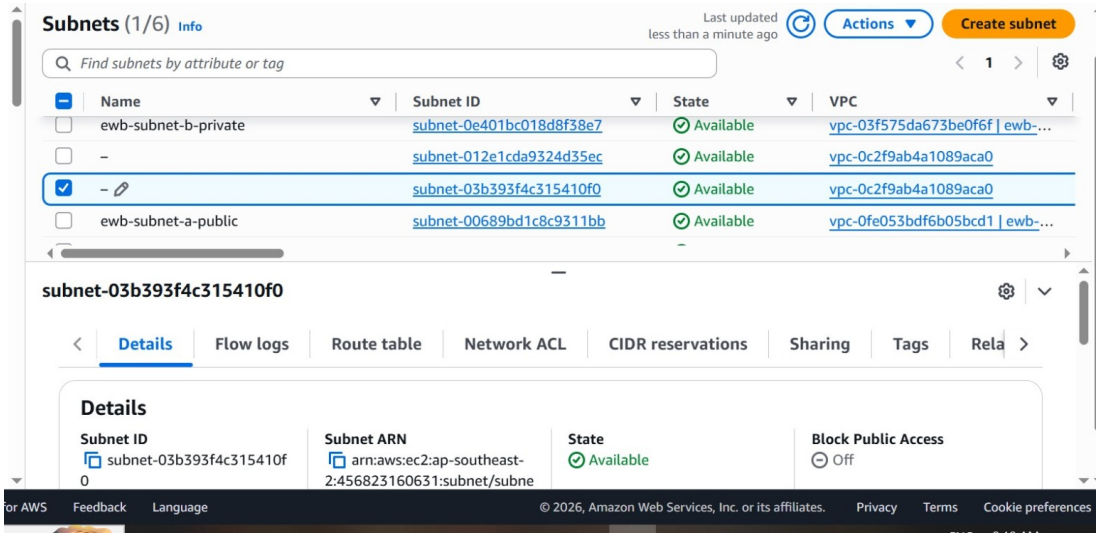


Figure 2. Subnets page showing available subnets, including ewb-subnet-a-public and ewb-subnet-b-private.

4. AWS Console Evidence (continued)

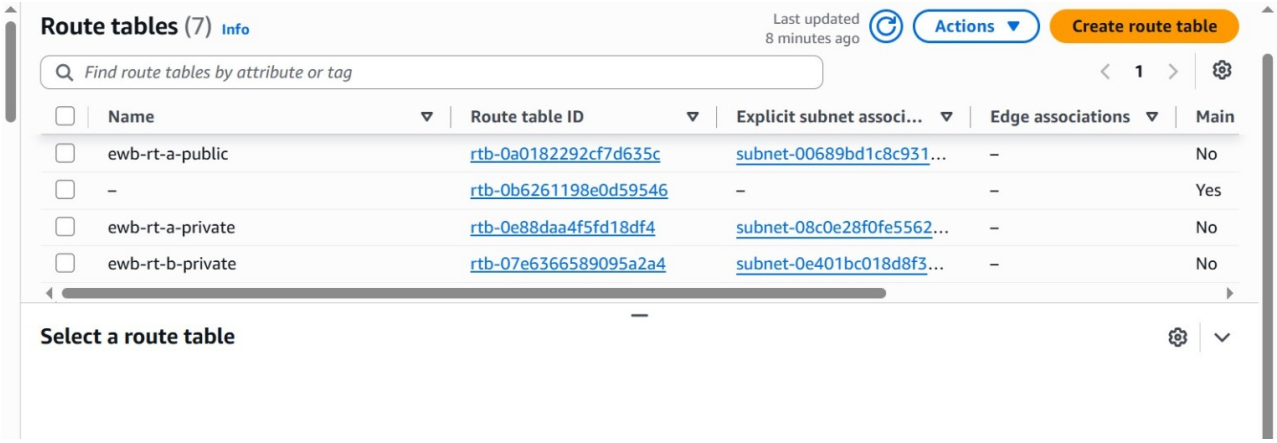


Figure 3. Route tables in the VPC console, including ewb-rt-a-public, ewb-rt-a-private, and ewb-rt-b-private.

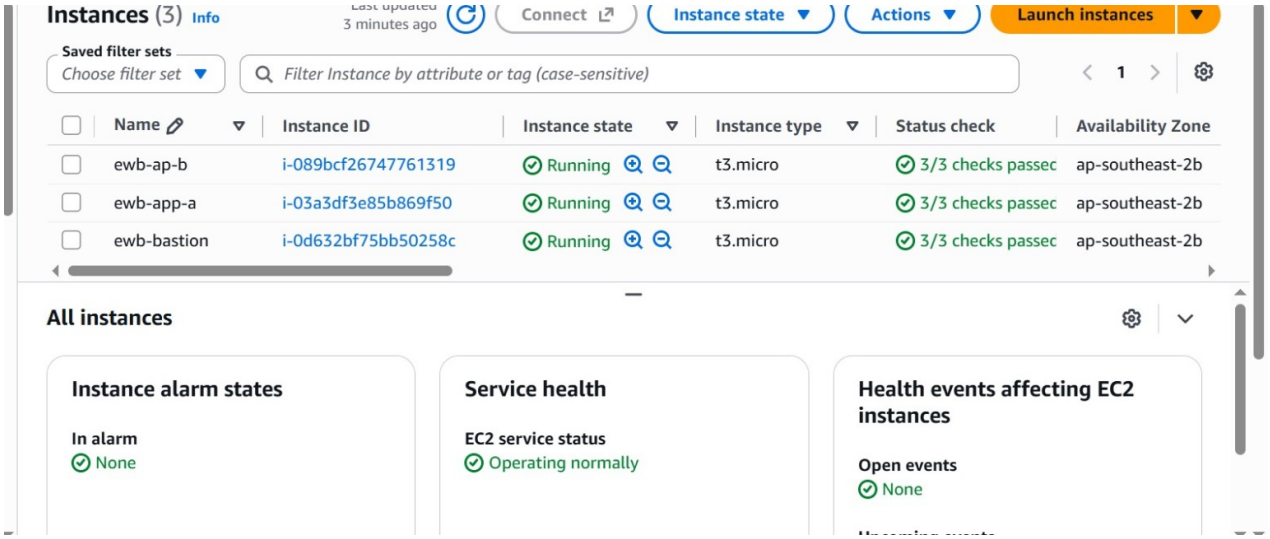


Figure 4. EC2 instance list showing ewb-ap-b, ewb-app-a, and ewb-bastion running, with status checks passed.

4. AWS Console Evidence (continued)

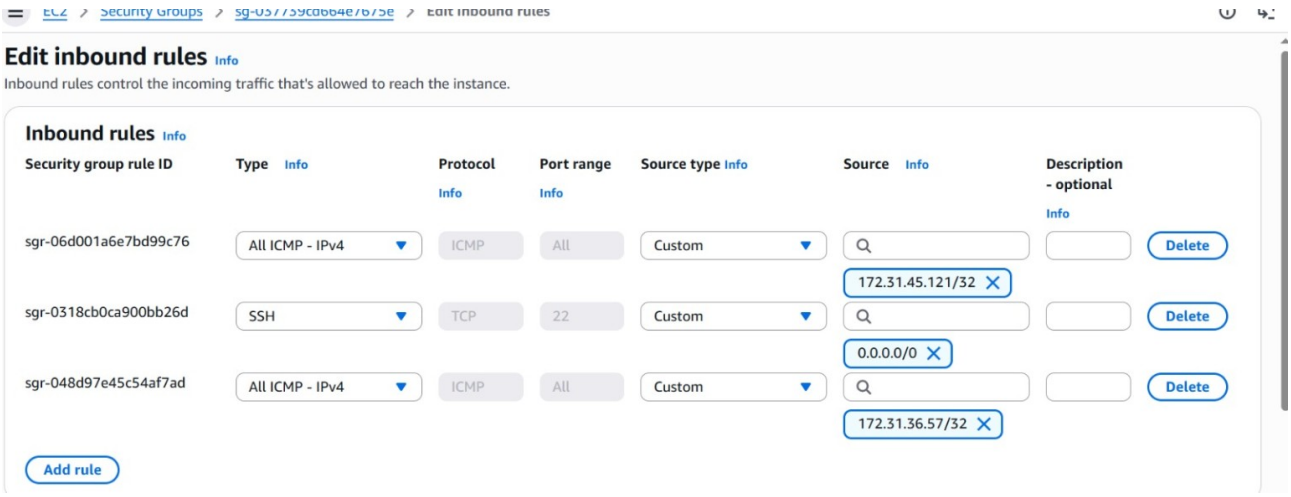


Figure 5. Inbound security group rules, including SSH and ICMP IPv4 sources used for the connectivity tests.

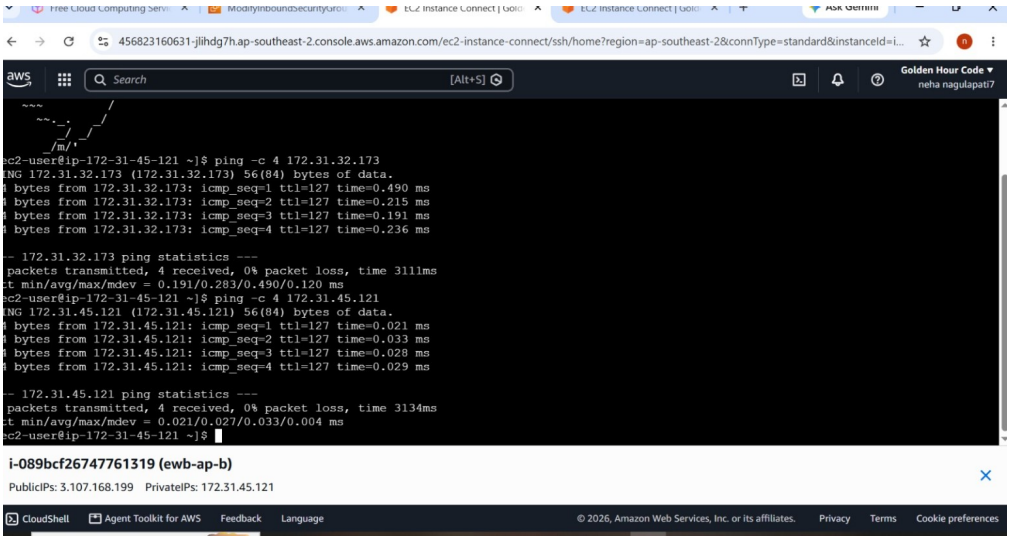


Figure 6. EC2 Instance Connect terminal showing successful ping tests between 172.31.45.121 and 172.31.32.173, with 0% packet loss.

5. Configuration and Testing Summary

The EC2 console screenshot shows all three instances in the Running state and passing their status checks. The security-group screenshot shows SSH access configured from 0.0.0.0/0 and ICMP IPv4 rules using specific private-IP sources. The terminal screenshot records successful ping replies between 172.31.45.121 and 172.31.32.173.

6. Results

The supplied terminal output shows four packets transmitted and four received for each displayed ping test, resulting in 0% packet loss. This confirms that the two shown private IP addresses were reachable from the terminal at the time of testing.

7. Security Note

For a real deployment, SSH should normally be restricted to a trusted administrator IP address or a suitable managed access method rather than being open to 0.0.0.0/0. ICMP rules should also be limited to the required source instances and removed when no longer needed.

8. Conclusion

This exercise provided practical exposure to AWS VPC resources, subnet and route-table inspection, EC2 instance status verification, security-group rules, and private-IP connectivity testing. The recorded ping output indicates successful communication between the two application-instance private IP addresses during the test.

Before submitting: fill in your name, college roll number, department/course, and college on the cover page. Review the report and make sure it matches the configuration in your AWS account.